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In [ ]: #STEP-1: Importing the Libraries
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In [2]: import pandas as pd
import matplotlib.pyplot as plt
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In [ ]: #STEP-2: Loading the data
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In [6]: df = pd.read_csv('Netflix_Data.csv')
print(df.head())
```

	Show_id	Type	Title	Director	\
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	
1	s2	TV Show	Blood & Water	NaN	
2	s3	TV Show	Ganglands	Julien Leclercq	
3	s4	TV Show	Jailbirds New Orleans	NaN	
4	s5	TV Show	Kota Factory	NaN	

	Cast	Country	\
0	NaN	United States	
1	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	
2	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	
3	NaN	NaN	
4	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	

	Date_added	Release_year	Rating	Duration	\
0	September 25, 2021	2020	PG-13	90 min	
1	September 24, 2021	2021	TV-MA	2 Seasons	
2	September 24, 2021	2021	TV-MA	1 Season	
3	September 24, 2021	2021	TV-MA	1 Season	
4	September 24, 2021	2021	TV-MA	2 Seasons	

	Listed_in	\
0	Documentaries	
1	International TV Shows, TV Dramas, TV Mysteries	
2	Crime TV Shows, International TV Shows, TV Act...	
3	Docuseries, Reality TV	
4	International TV Shows, Romantic TV Shows, TV ...	

	Description
0	As her father nears the end of his life, filmm...
1	After crossing paths at a party, a Cape Town t...
2	To protect his family from a powerful drug lor...
3	Feuds, flirtations and toilet talk go down amo...
4	In a city of coaching centers known to train I...

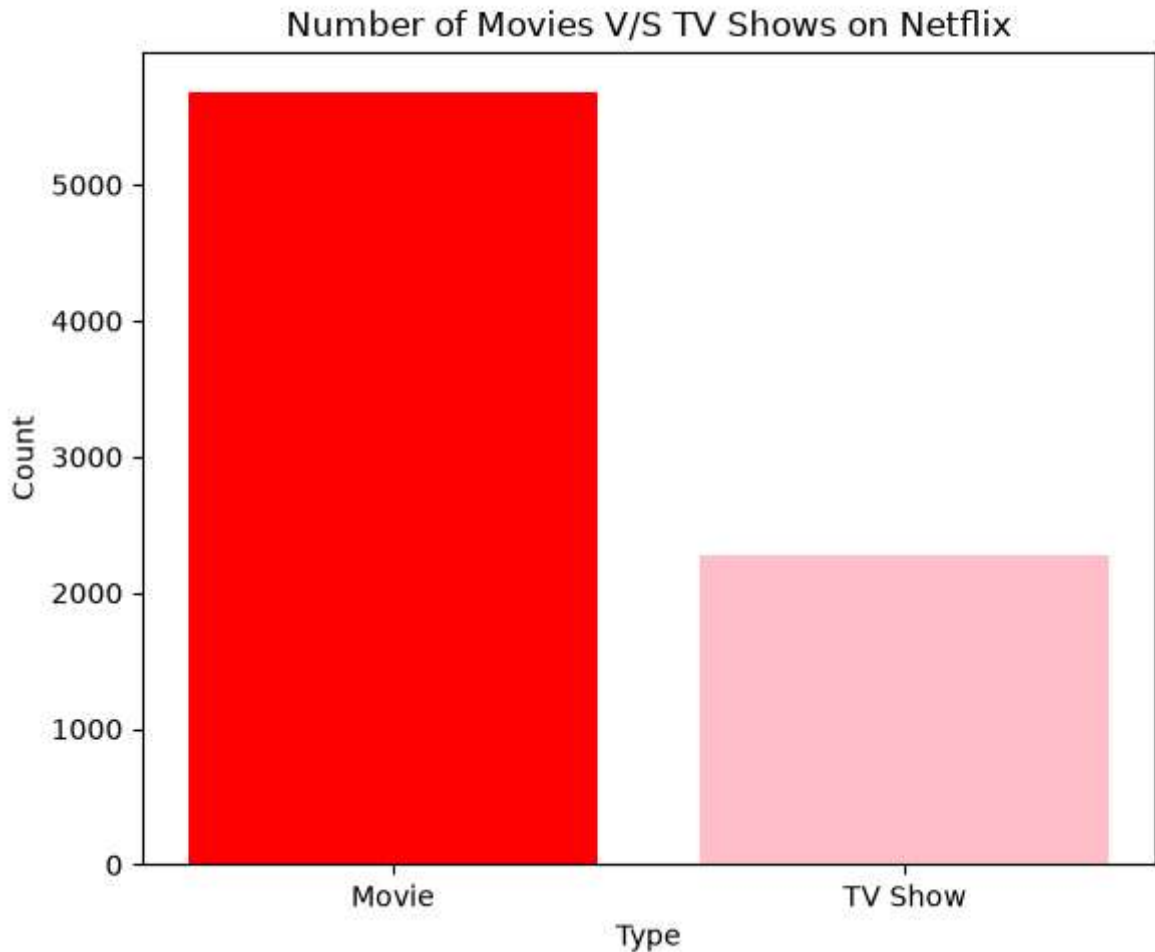
```
In [6]: #STEP-3: Cleaning the data
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In [7]: df = df.dropna(subset=['Type', 'Release_year', 'Rating', 'Country', 'Duration'])
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In [ ]: #STEP-4: Visualization
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In [58]: '''
BAR CHART:- Movies V/S TV shows
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'''
type_counts = df['Type'].value_counts()
plt.figure(figsize = (6,5))
plt.bar(type_counts.index, type_counts.values, color=['red', 'pink'])
plt.title('Number of Movies V/S TV Shows on Netflix')
plt.xlabel('Type')
plt.ylabel('Count')
plt.tight_layout()
plt.savefig('Movies_vs_TVshows.png')
plt.show()
```

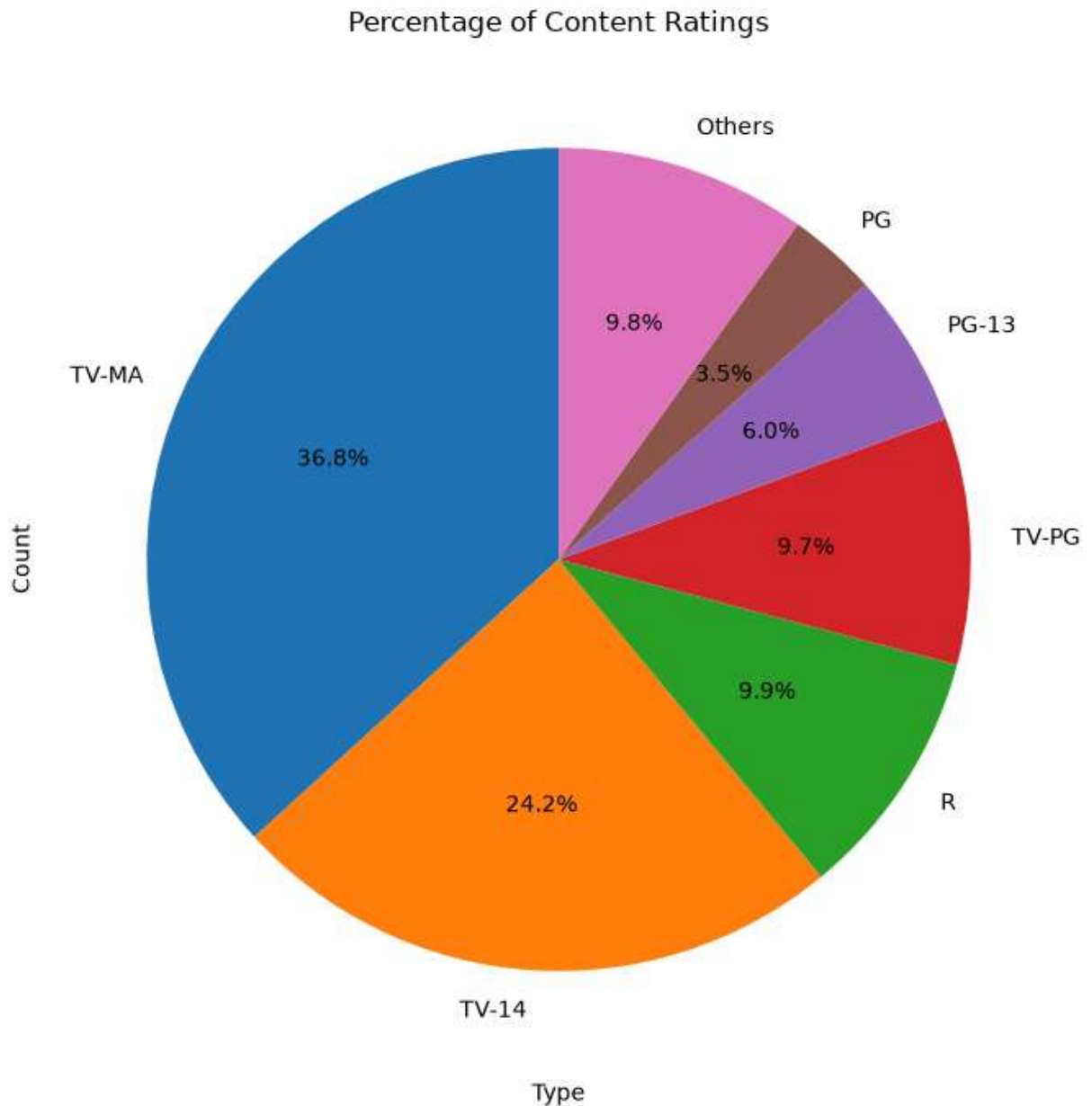


```
In [ ]: '''Conclusion'''
Movies are much more numerous than TV Shows in the Netflix dataset.
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In [ ]:
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In [42]: '''
PIE CHART:- Content Rating Distribution
'''
rating_counts = df['Rating'].value_counts()
top = rating_counts[:6]
others = rating_counts[6:].sum()
top["Others"] = others
plt.figure(figsize = (7,7))
plt.pie(top, labels = top.index, autopct = '%1.1f%%', startangle = 90)
plt.title('Percentage of Content Ratings')
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plt.xlabel('Type')
plt.ylabel('Count')
plt.tight_layout()
plt.savefig('Content_Ratings_pie.png')
plt.show()
```

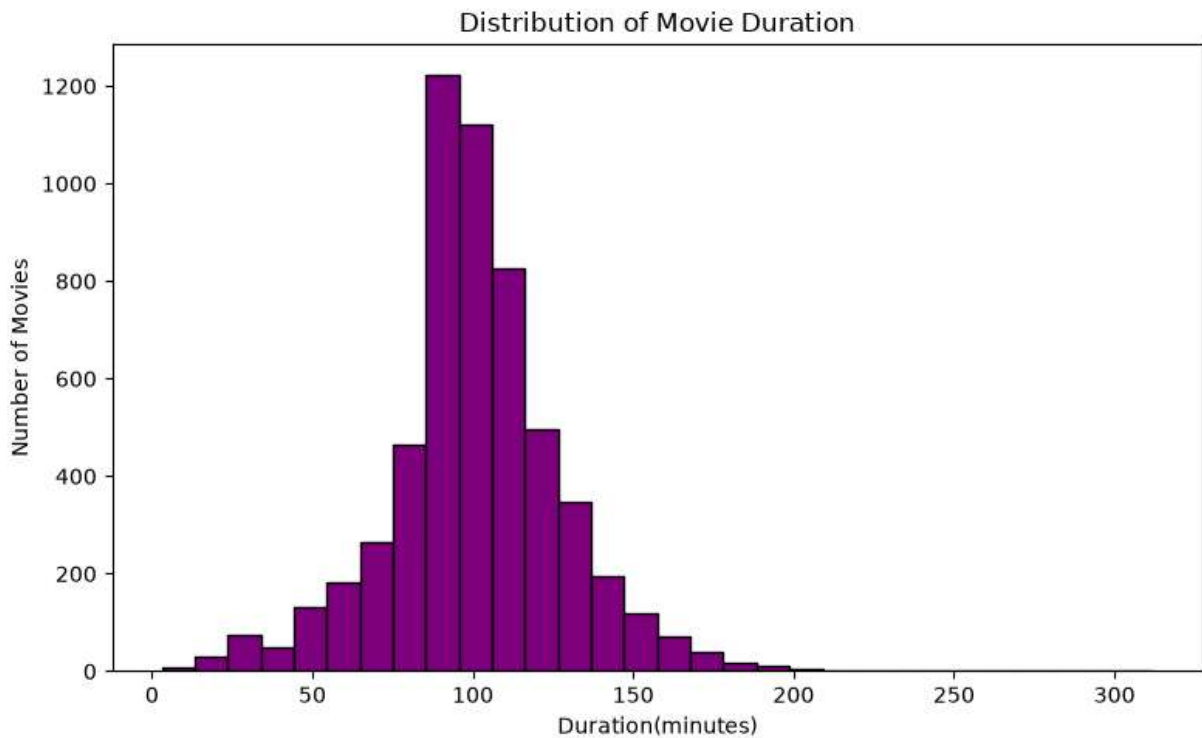


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In [ ]: '''Conclusion'''
TV-MA(36.8%) is the largest category while PG-13(6.0%) and PG(3.5%)
accounts for smaller shares. Other category combines (9.8%) in total.
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In [ ]:
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In [57]: '''
HISTOGRAMS:- Movie Duration Distribution
'''
movie_df = df[df['Type'] == 'Movie'].copy()
movie_df['duration_int'] = movie_df['Duration'].str.replace('min', '').astype(int)
```

```
plt.figure(figsize=(8,5))
plt.hist(movie_df['duration_int'], bins=30, color='purple', edgecolor = 'black')
plt.title('Distribution of Movie Duration')
plt.xlabel('Duration(minutes)')
plt.ylabel('Number of Movies')
plt.tight_layout()
plt.savefig('Movie_Duration_histogram.png')
plt.show()
```

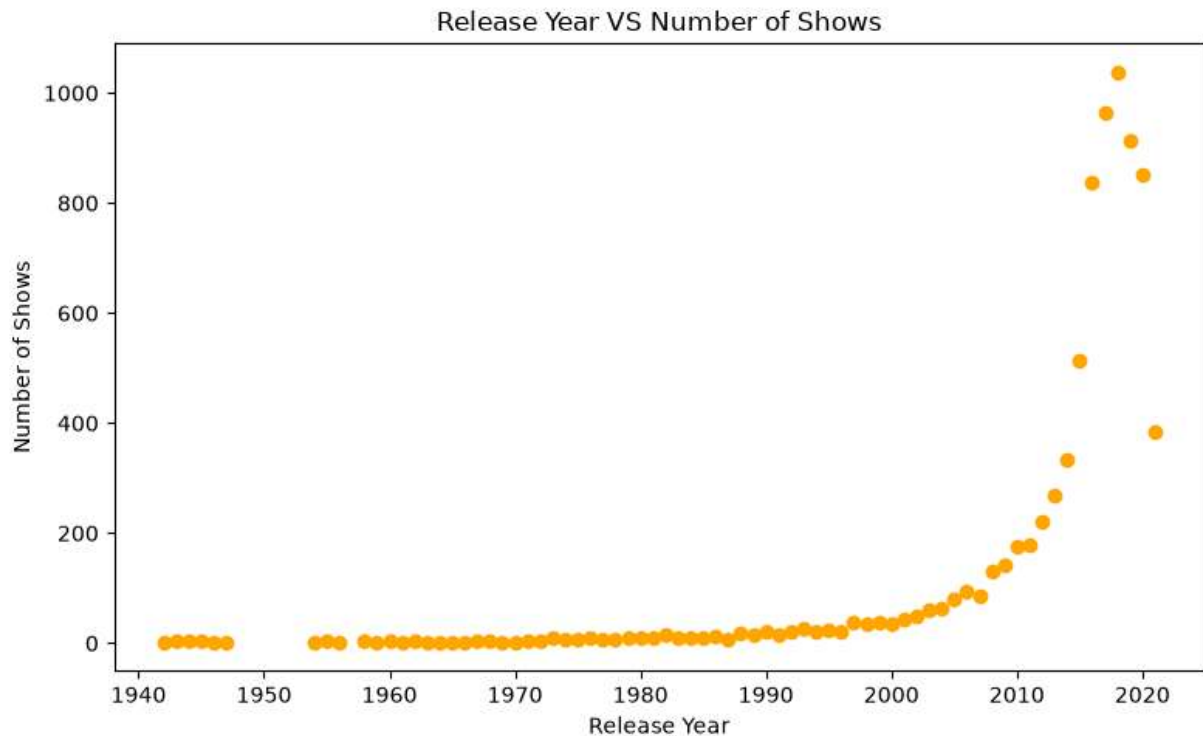


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In [ ]: '''Conclusion'''
Most Netflix movies are concentrated around 90-120 minutes
which indicates Netflix primarily hosts full-length movies.
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In [ ]:
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In [ ]: '''
SCATTER PLOT:- Release Year V/S Number of Shows
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In [55]: release_counts = df['Release_year'].value_counts().sort_index()
plt.figure(figsize=(8,5))
plt.scatter(release_counts.index, release_counts.values, color = 'orange')
plt.title('Release Year VS Number of Shows')
plt.xlabel('Release Year')
plt.ylabel('Number of Shows')
plt.tight_layout()
plt.savefig('Release_Year_scatter.png')
plt.show()
```



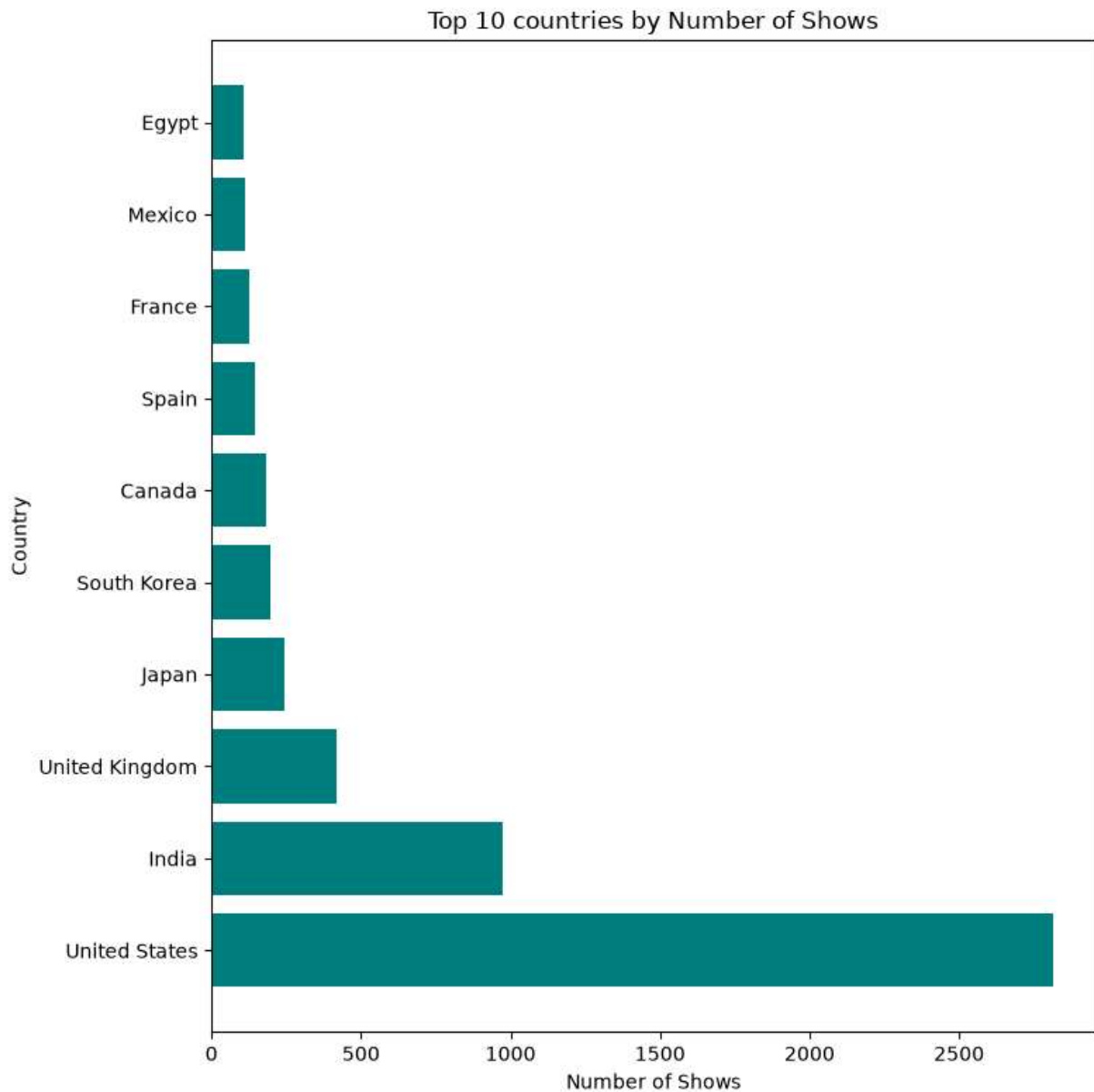
In []: `'''Conclusion'''`

The number of shows increases significantly in more recent years, indicating that Netflix content library expanded rapidly before slowing in the latest years.

In []:

In []: `'''`
 BAR GRAPH: Top 10 countries shows
`'''`

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In [65]: country_counts = df['Country'].value_counts().head(10)
plt.figure(figsize=(8,8))
plt.barh(country_counts.index, country_counts.values, color = 'teal')
plt.title('Top 10 countries by Number of Shows')
plt.xlabel('Number of Shows')
plt.ylabel('Country')
plt.tight_layout()
plt.savefig('Top 10_Countries.png')
plt.show()
```



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In [ ]: '''Conclusion'''
        The United States has the highest number of Netflix title, dominating the catalogue
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In [ ]:
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In [ ]: '''
        SUBPLOT:- Movie V/S TV Shows by Year
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```
In [12]: content_by_year = df.groupby(['Release_year', 'Type']).size().unstack().fillna(0)

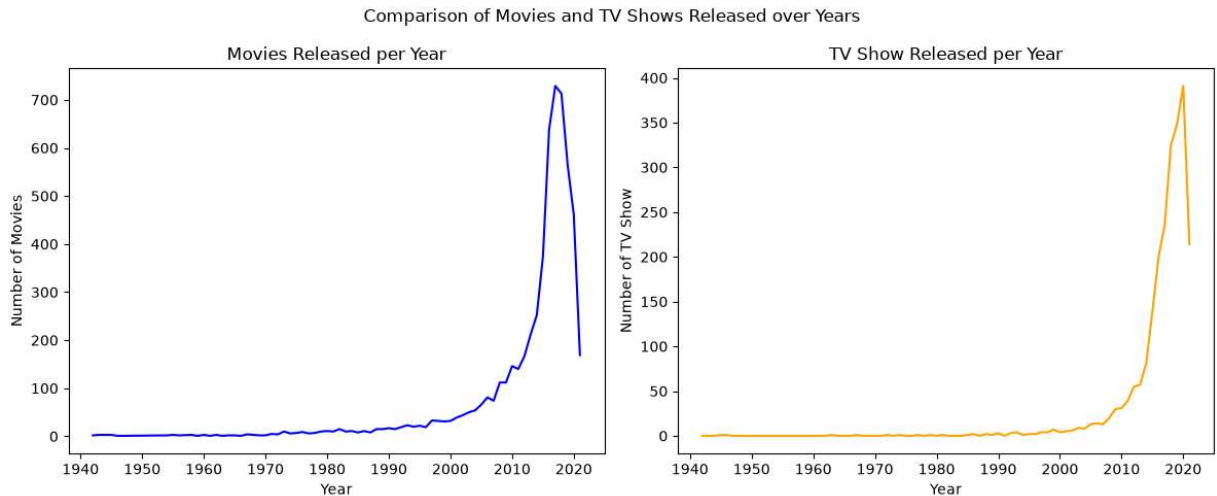
fig, ax = plt.subplots(1,2, figsize=(12,5))

#First subplot:movies
ax[0].plot(content_by_year.index, content_by_year['Movie'], color = 'blue')
ax[0].set_title('Movies Released per Year')
ax[0].set_xlabel('Year')
ax[0].set_ylabel('Number of Movies')
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```
#Second subplot:TV shows
ax[1].plot(content_by_year.index, content_by_year['TV Show'], color = 'orange')
ax[1].set_title('TV Show Released per Year')
ax[1].set_xlabel('Year')
ax[1].set_ylabel('Number of TV Show')

fig.suptitle('Comparison of Movies and TV Shows Released over Years')

plt.tight_layout()
plt.savefig('Movie_TV_Shows_Comparison.png')
plt.show()
```



```
In [ ]: '''Conclusion'''
Both show an increasing trend over the years, with a significant rise in
recent years, indicating Netflix rapid content expansion. Movies consistently
outnumber TV Shows, showing that Netflix library is primarily movie-based
while TV Shows have also experienced strong growth
```